Ayush Agarwal

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Summary!

Multidisciplinary Data Scientist and Software Engineer with a strong foundation in AI/ML, predictive modeling, and full-stack development. Proficient in Python, Java, and React, with proven experience building end-to-end data products—from geospatial data engineering to deploying interactive ROI dashboards. Adept at combining technical depth in algorithms (XGBoost, Deep Learning) with strategic business analysis to deliver scalable, actionable solutions.

Education

Vellore Institute of Technology(Chennai Campus), B.Tech in Computer Science and Engineering with specialization in AI & ML — Chennai, India — Sept 2022 – Present

- CGPA: 8.72

Narayana E-Techno School,12th — Mumbai, India — May 2020 – May 2022

- Grade: 88%

RBK School, 10th — Mumbai, India — Sept 2022 – Present

- Grade: 94%

Certifications

Artificial Intelligence using TensorFlow, SmartInternz / Google — Jun 2024 – Jul 2024

Skills & Interests

Programming Languages: Python, Java, JavaScript, HTML/CSS, PostgreSQL

Frameworks: pandas, numpy, scikit-learn, matplotlib, seaborn, YOLOv8, React

Tools & Platforms: Tableau, AWS, Jupyter, Google Colab, Git, VS Code

Computer Science Fundamentals: Data Structures & Algorithms (Trees, Graphs), REST APIs, SDLC, Agile

Soft Skills: Problem Solving, Team Collaboration, Quick Learning, Communication

Interests: Table Tennis, Creative Writing, Reading psychology

Projects

InsightIQ: Autonomous Text-to-SQL Agent — GitHub | Live Demo

- Architected a RAG-based BI agent using Gemini and ChromaDB to query SQL databases via natural language; implemented an agentic self-correction loop that autonomously fixes syntax errors.
- Developed a full-stack Streamlit dashboard with automated dynamic visualization (Bar/Line charts) and deployed with role-based access control, reducing ad-hoc reporting time by 80%.
- Tools Used: Python, Gemini, LangChain, RAG, ChromaDB, Streamlit, Pandas

SCM-Optima: AI Supply Chain Control Tower — GitHub | Live Demo

- Developed an end-to-end analytics platform processing 180,000+ orders; engineered a Random Forest classifier (83% accuracy) to predict shipment delays and enable proactive risk mitigation.
- Designed a dynamic inventory optimization engine using Prophet for 30-day demand forecasting and automated Safety Stock/Reorder Point calculations, displayed via a real-time Streamlit dashboard.
- Tools Used: Python, Streamlit, Prophet, Scikit-Learn, Plotly, Pandas

VCheck: Uniform Compliance System

- Engineered an AI model using YOLOv8 to detect uniform compliance from images with 98% accuracy. Trained on custom dataset to automate visual inspection.

NYC Rental Arbitrage Engine GitHub | Live Demo

- Engineered an automated pipeline fusing 11,000+ Airbnb listings with Zillow data using Geopandas; built an XGBoost price prediction model ($R^2 = 0.51$) interpreted via SHAP analysis.
- Designed a Cash-on-Cash ROI calculator identifying >8% yield opportunities and deployed a live Streamlit dashboard featuring interactive choropleth maps for real-time filtering.
- Tools Used: Python, Geopandas, XGBoost, SHAP, Streamlit, Folium

Digital Farming (IoT)

- Designed a hydroponics-based farming system with IoT sensors for real-time soil moisture and crop health tracking. Integrated alert systems and dashboard.

Experience

Web Developer Intern, Afame Technologies — Remote Apr 2024 – May 2024

- Developed and optimized dynamic, responsive web pages using HTML, CSS, and JavaScript.
- Improved load times and UX for client websites.

Publications

Predictive Modelling of Physicochemical Properties of Cissus Quadrangularis Compounds

Journal: Letters in Applied NanoBioScience
Built regression models (Random Forest, AdaBoost, Linear) to model compound behaviors.
10.33263/LIANBS143.192

Looking-Glass Upon the Wall... Body Image and Social Media's Impact on Indian Youth (Under Review)

Journal: Indian Journal of Social Psychiatry
Led data visualization and statistical analysis for study involving 143 students; explored body image and self-esteem through ML-backed insights.

Quantitative Structure-Property Relationship (QSPR) Analysis of Cissus Quadrangularis Compounds Using Topological Indices and Machine Learning Models (Under Review)

Journal: Journal of Healthcare Informatics Research
Applied ML algorithms to predict compound properties using topological descriptors.

Modelling of Physicochemical Properties of Papaya Leaf Compounds Using Topological Indices and Random Forest Regression (Submitted)

Journal: Journal of Applied Statistics
Developed predictive models using topological indices to estimate compound properties via Random Forest regression.

Predictive QSAR Modelling of Bioactivity for Phytochemicals from Carica papaya: A Comparative Analysis of Machine Learning Regression Algorithms on a Pilot Dataset(Submitted)

Journal: Polycyclic Aromatic Compounds
Trained and compared multiple Regression models for predicting the bioactivity for Phytochemicals extracted from Papaya.

Proposing the Urban Plant Suitability Index (UPSI): A Multi-Criteria Framework for Selecting Flora in Tropical Urban Ecosystems(Submitted)

Journal: Forest Ecology and Management
Proposed a Urban Plant Suitability Index(UPSI) for urban planning and boost urban forestry.